

Statistical Screening for Poisoned Machine-Learning Training Data

Detection and Quarantine of Suspicious Samples at the Data-Ingestion Boundary

github.com/poojakira/dataset-poisoning-detector - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

199

Tests passed

90.88%

Statement coverage

0.55

Cross-class F1 @ 5% poison

0.595

Cross-class F1 @ 10% poison

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface

Data, models, requests, policies, or artifacts entering the tested path.

Control surface

Repository-specific validation, analysis, and fail-closed checks.

Evidence surface

Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01

Validate incoming feature batches and streaming samples.

02

Apply z-score, IQR, Isolation Forest, and spectral/label-aware detectors.

03

Combine detector signals for screening and quarantine workflows.

04

Maintain streaming state and deduplication metadata.

05

Benchmark label-flip behavior with committed reproducible artifacts.

Research Contribution

An ingestion-boundary screening pipeline combining statistical outlier methods with a label-aware cross-class centroid detector that improves the committed synthetic label-flip benchmark from the old top-1 spectral baseline 0.08 / 0.23 / 0.3675 F1 to 0.55 / 0.595 / 0.6975 F1 at 5% / 10% / 20% contamination.

Verified Evidence

EV-01 199 tests passed, 0 skipped.

EV-02 90.88% statement coverage; CI gate is 90%.

EV-03 Ruff check/format and Pyright pass on the final branch; GitHub currently reports 0 open Code Scanning, Dependabot, and secret-scanning alerts.

EV-04 Committed results/spectral_benchmark.json records cross-class centroid F1:

EV-05 0.55 at 5% contamination

EV-06 Cross-class F1 @ 20% poison: 0.6975

Method and Control Path

01 Validate incoming feature batches and streaming samples.

02 Apply z-score, IQR, Isolation Forest, and spectral/label-aware detectors.

03 Combine detector signals for screening and quarantine workflows.

04 Maintain streaming state and deduplication metadata.

05 Benchmark label-flip behavior with committed reproducible artifacts.

Limitations and Residual Risk

- The benchmark is synthetic and separable; it is not proof of production detection rate.
- Clean-label, subtle backdoor, and very-low-rate poisoning remain difficult.
- Throughput is hardware/environment dependent; no fixed product SLA is claimed.
- Stateful streaming behavior requires deliberate partitioning/external state for horizontal scaling.

Reproducibility

Expected current evidence: 199 passed, 90.88% coverage.

Evidence Provenance

Claim discipline

Metrics come from the repository's verified metrics and research brief.

Scope discipline

Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction

PDF and PNG are generated locally from committed evidence without paid services.